

A Review of Multi-Sensor Fusion System for Large Heavy Vehicles Off Road in Industrial Environments

De Jong Yeong
Dept. of Technology, Engineering and Maths
Institute of Technology, Tralee
Tralee, Ireland
de.jong.yeong@research.ittralee.ie

Dr. John Barry
STEM, IMaR Research Centre
Institute of Technology, Tralee
Tralee, Ireland
john.barry@staff.ittralee.ie

Prof. Joseph Walsh
STEM, IMaR Research Centre
Institute of Technology, Tralee
Tralee, Ireland
joseph.walsh@staff.ittralee.ie

Abstract—Industry 4.0 or fourth industrial revolution elevates the computerization of Industry 3.0 and enhances it with smart and autonomous systems driven by data and Machine Learning. This paper reviews the advantages and disadvantages of sensors and the architecture of multi-sensor setup for object detection. Here we consider the case of autonomous systems in for large heavy vehicles off-road in industrial environments with the use of camera sensor, LiDAR sensor, and radar sensor. Understanding the vehicles surroundings is a vital task in autonomous operation where personnel and other obstacles present significant hazard of collision. This paper review further discusses the challenges of time synchronisation on sensor data acquisition in multi-modal sensor fusion for personnel and object detection, and details a solution implemented in a Python environment.

Keywords—sensor fusion, object detection, autonomous systems, multi-sensor, time synchronisation.

I. INTRODUCTION

The word “autonomous” from the Cambridge Dictionary refers to the ability of an autonomous system to operate in a real world environment without being monitored for extended periods of time [1]. The development of technologies associated with autonomous vehicles has advanced at a significant pace over the last number of years. The SAE International summarised six distinct levels of vehicles automation from level 0 where no autonomous features available on vehicles to level 5 where vehicles are able to operate without human supervision [2]. It is indispensable for autonomous vehicles to perceive their surroundings accurately based on reliable object detection and classification in order to perform any subsequent actions safely. A variety of sensors are used for this purpose.

In this paper we describe initial efforts to synchronously acquire data from three different types of sensors – camera, LiDAR and radar. Each type of sensor has specific advantages and disadvantages such as the ability to resolve colour information (camera), provide accurate depth information (Lidar) and be insensitive to rainfall and ambient lighting levels (radar) [3]. Current driver assistance systems available in production, such as Blind Spot Monitoring (BSM), and Active Cruise Control (ACC) are usually independent systems, where functionalities and sensors of a system are located in one electronic control unit (ECU). For example, most ACC applications are based only on radar sensor. However, data from multiple sensors are required to be fused in order to build a

consistent model that is capable to perceive their surroundings accurately in a range of environments. The technique of fusing data acquired from multiple sensors is called sensor fusion [4]. Sensor fusion is the process of integrating data from multiple sensors to reduce the amount of uncertainty as compared to when sensors are used individually [5].

High-level fusion (HLF) is a very popular method for fusing multiple sensors data at present, compared with low-level fusion (LLF). In HLF approach, objects are detected with each sensor separately and subsequently detections are combined. However, the drawback of HLF is that it provides inadequate information as classifications with low confidence value are discarded, if for example, there are several overlapping objects. In contrast, with LLF approach, data from different sensor types are combined at raw data level, thus all information is retained and this potentially improves the accuracy of object detection. LLF is complex and comes with a multitude of challenges. An extremely accurate extrinsic calibration of sensors is required to precisely fuse their perceptions of the environment in order to perform subsequent actions safely. Additionally, the sensor recordings are required to be time-synchronised and counterbalanced of ego-motion [6].

Modern driver assistance systems incorporate several different sensors whose data must be merged. Data association is the first part in sensor fusion algorithms which are otherwise unable to provide precise timestamp up to millisecond accuracy. Data association is a method to associate the measurements to targets with certainty when several targets are in the same neighbourhood, which is the central problem in multi-sensor fusion for multiple object detection [7]. Inaccurate timestamps between sensor data may result in, for example, incorrect estimation of the positions of detected objects. Measurements taken by sensors in general do not include timestamps, consequently the measurements are only timestamped after they are arrived at a common data acquisition system. However, the immeasurable latencies before the arrival of measurements at the common data acquisition system cannot be determined directly. It is crucial not to overlook these unknown latencies as they may total up to several hundred milliseconds, which is not acceptable, particularly in time critical situations, e.g. pre-crash situations [8-9].

The objective of the present work is to develop a sensor system and sensor fusion architecture, with the initial focus on a means of synchronising the data acquired from each sensor. An overview of the prior art is presented in Section II. In Section

The research reported in this paper is supported, in part by Science Foundation Ireland Grant 13/RC/2094 and co-funded under the European Regional Development Fund through the Southern & Eastern Regional Operational Programme to LERO, The Science Foundation Ireland Research Centre of Software.

III, we discuss the pros and cons of various sensors. In Section IV, we discuss our pre-experimental findings and challenges of time synchronisation in our multi-sensor setup for large heavy off-road vehicles. Finally, we present a summary and an outlook of future work.

II. RELATED WORK

Most of the recent research generally focuses on a high-level fusion approach or a multi-sensor integration approach for object detection in autonomous vehicles [10]. Multi-sensor integration differs from sensor fusion in that sensor fusion incorporates the actual combination of sensory information from several sensors into one representational format [11]. Kim, S. et al. developed a high-level multi-sensory integration system with the use of laser scanner and stereo vision for object matching between sensors, for on-road vehicle detection. The authors also designed a synchronisation process to address time latencies between two sensors, object age to track results from both sensors, and a re-ordering algorithm for laser scanner data for robust tracking of the objects. Their experimental results demonstrated an accuracy of 99% for several manoeuvring cases including downtown, city and highway roads [10]. The work by Jahromi, et al. [12] is based on environmental perception using multi-sensor fusion framework for autonomous vehicles with 8 radar sensors, a LiDAR sensor and 8 cameras to provide rich sensory data. The proposed Fully Convolutional Network (FCNx) model and the implementation of Extended Kalman Filter provides for detecting and tracking objects on the road as well as providing information on safe drivable road region or zones. The results demonstrated a Cross Entropy Loss of 3.6% (a perfect model would have a loss of zero percent) and a Root Mean Square Error (RMSE) of less than 0.1.

Our approach to sensor system design builds upon the work by Jahromi, et al. [12] and Kocić, et al. [28] to detect and track objects and will incorporate a means of synchronising the data acquired from each system. A camera, a LiDAR sensor and a radar sensor is used in our present work relating to sensor systems in large heavy vehicles off-road.

With regards to time synchronisation in multi-sensor fusion systems: This is vital to perform complex sensing tasks like perceiving environments in real time, and is categorised into internal synchronisation and external synchronisation. Internal synchronisation refers to where clocks are synchronised without support of a reference clock to obtain a consistent view of time across all nodes in the sensor network. External synchronisation refers to where clocks are synchronised with an external source of time (reference clock) that is precisely relatable to a real-time standard such as UTC [26]. Unknown latencies that cannot be determined directly are one of the challenges in time synchronisation. Measurements may arrive at a sensor data acquisition system after a certain unknown delay and may be caused by several factors such as computer scheduling time, pre-processing time, communication transfer delays. Additionally, detecting the occurrence of frame drops is another time-related challenge to resolve [8]. The first phase of our research evaluates the potential to synchronise sensor data where external reference clocks are not conveniently accessible.

III. SENSORS

This section discusses the benefits and limitations of the sensors used in our proposed sensor system. Here we consider a Falcon-IQ Internet Protocol (IP) industrial zoom camera of the model EZIP-T030(E), a Velodyne LiDAR Ultra Puck VLP-32C sensor, and a SmartMicro automotive radar UMRR-96 Type 153 sensor in our proposed systems. The arrangement of multiple sensors on large heavy vehicles off-road is shown in Fig. 1.



Fig. 1. Parallel arrangement of multi-sensor (camera, LiDAR and Radar) used for environment perception in autonomous vehicles for large heavy vehicles off-road in industrial environments.

A. Camera

Cameras are generally reliable and relatively inexpensive. Vehicle applications including advanced driver assistance systems rely on cameras to perform high resolution tasks such as environmental perception, object classifications and colour perception. The output of a conventional 'RGB' camera is typically an array of several hundred by several hundred (or more) 'pixels', each pixel having a red, green and blue intensity level [13]. Image noise may include periodic moire-type patterns cause by imperfect lenses, filters and image sensors [12] and most camera systems will have some degree of image distortion which may require calibration [29-30].

Conventional monocular cameras usually provide high-resolution videos and/or images but are passive in that they lack native depth information and are dependent on suitable illumination and environmental conditions. Poor environmental conditions such as strong contrast, glare, fog, snowfall and heavy rain may produce lower quality images for which object detection algorithms may prove less reliable. Nevertheless, the colour perception provided by a camera is a critical contribution to the overall sensor systems architecture [14-15]. The camera used in our proposed sensor system has a pixel array (image resolution) of 1920x1080 with a variable field of view (FOV) of between approximately two to 65 degrees included angle. Configuration settings of the camera like FOV, calibration etc. are required before camera data acquisition process.

B. LiDAR Sensor

LiDAR stands for Light Detection and Ranging. LiDAR works on the principle of emitting laser or infrared light pulses using rotating lenses and measuring the duration of time to register reflections of the individual light pulses. From the time of flight and the speed of light, a distance map of the

surrounding environment may be established. The wavelength of LiDAR sensors used in autonomous vehicles are commonly 905 nm and are capable of performing well in fog or light rain, particularly with longer wavelength light [12]. The sensor uses an active illumination sensor that is able to collect data in various lighting condition [17]. An experiment conducted by Kutila, M. et al. demonstrated that weather conditions such as fog, snow or rain will degrade the performance of sensor by 25% [18].

LiDAR sensors may also provide various signal intensity characteristics depending on the nature of the perceived objects. The point cloud data from LiDAR sensors can be used to generate a full 3D spatial representation (or laser images) of the environment around the autonomous vehicle. Different LiDAR sensors have different specifications, and understanding the significance of each specification is critical in choosing the most suitable LiDAR for specific applications like mapping features under forest canopy or obstacle detection [16].

LiDAR sensors permit fast data acquisition and processing and provide high accuracy and resolution. LiDAR sensors do not provide colour information of the perceived surroundings, as do RGB cameras. The LiDAR used in our proposed sensor system provides a horizontal view of 360° that can be configured and a vertical field of view of 40°. It reportedly combines a measurement range of 200m with a spatial and depth resolution of 0.1° to 0.4° and $\pm 3\text{cm}$ respectively [19].

C. Radar Sensor

Radar, which stands for Radio Detection and Ranging was first developed before World War II and has been used to identify the direction, speed and position of moving objects for decades. It works by transmitting electromagnetic (EM) radio waves at a target area and monitoring for reflections from any ‘illuminated’ objects. There are three major classes of automotive radar systems: short-range radar for parking assist systems; medium-range radar for side or rear collision avoidance systems and blind-spot detection; long-range radar for early collision detection [20]. Long-range radar requires higher power amplification for the signals. Radar sensors are usually impervious to adverse weather conditions like heavy rain, fog weather, snowfall etc. and function irrespective of lighting conditions. Radars usually operate at frequencies of 24 Gigahertz (GHz), 77GHz and 79 GHz [12]. The current commercial 24GHz radar sensors have more limited resolution which can lead to problems in identifying and reacting to multiple hazards [20]. Radars are capable of directly measuring the velocity of objects with high precision, however, measuring the position of objects due to limited resolution and detecting static objects remains a challenge even after significant innovation in the field of Imaging Radar [15].

The radar used in our proposed sensor systems operates in a frequency of 77GHz and offers an azimuth field of view of either 100° or 130° and an elevation field of view of 15°. Different classes of automotive radar systems (long-range; medium-range; short-range) offer distinct minimum and maximum target range values but may vary depending on environmental ‘clutter’.

IV. FINDINGS / RESULTS

This section discusses the initial experimental findings and challenges relating to time synchronization in our proposed multi-sensor setup for large heavy off-road vehicles, the components of which as shown in Fig. 2.



Fig. 2. Components of the proposed multi-sensor setup for large heavy vehicles off-road in industrial environments.

A Power over Ethernet (PoE) adapter along with a single Ethernet cable was used to provide both data connection and electric power to camera. A network switch is utilised to channel incoming sensor data from multiple input ports (LiDAR sensor and camera) to a specific output port that will take it toward its intended destination (PC). Besides, a CAN-USB adapter is used to enable connection between radar sensor and PC. A number of trials are conducted in lab to collect sensory data simultaneously afore setting multi-sensor systems on large heavy vehicles.

A. Simultaneous Sensor Data Acquisition

A pre-experimental solution was proposed to simultaneously collect multi-sensor data through implementation of algorithms in Python environment. Camera data was collected with the use of Open Source Computer Vision (CV) framework. This is a library of programming functions that are primarily aimed at real-time computer vision [21]. It is vital to ensure that the camera device time is synchronised with current local date and time or personal computer time before acquiring camera data.

A possible solution to resolve issues regarding to the device time synchronisation is to overlay current local date and time label from Python datetime module on each frame before outputting frames as Audio Video Interleave (AVI) file. The overlaid local date and time label can be retrieved using Python-Tesseract tool. The Python-Tesseract is an Optical Character Recognition (OCR) tool for Python. It is able to detect text content on images and translations of images to encoded text. However, the process of detecting text content on images may not be precise and requires human intervention to fix elements that were not correctly detected [22]. Pseudocode to acquire camera data and to detect datetime label on images is shown in Algorithm 1.

Algorithm 1 Acquire Camera Data and Detect Datetime Label on Images

```

1: initialize VideoCapture
2: initialize VideoWriter
3: while VideoCapture is open:
4:   read frame
5:   if read frame is TRUE then
6:     Overlay current local date and time label on frame
7:     VideoWriter output frame into AVI file
8:   end if
9: end while
10: release VideoCapture
11: release VideoWriter
12: VideoCapture read AVI file
13: for each frame (image) in AVI file:
14:   scale frame
15:   convert to grayscale
16:   threshold frame
17:   morphology noise removal
18:   invert black or white
19:   Python-Tesseract detect text content on frame
20:   save detected text to file
21: end for

```

Lidar data were acquired using Scapy, a Python module that enables the developer to acquire, sniff, forge and decode network data packets [23]. An algorithm was implemented with the used of pre-written method in Scapy framework to sniff and save raw data packets into a single .pcap file. VeloView, a third-party software developed by Velodyne is utilised to decode pcap files which store raw data packets into multiple comma-separated values (.csv) files for subsequent data analysis. The total number of .csv files represents the total frames captured throughout the duration of data acquiring process.

The algorithm for acquiring radar data works similarly to the algorithm implemented for acquiring lidar data. A Python-CAN module was utilized to collect raw radar data packets. It provides a suite of functionalities for sending and collecting messages on a Controller Area Network (CAN) bus [24]. A pre-written script provided by the module was used to capture and save raw radar data packets into a single .asc file. DriveRecorder3, a third-party software developed by SmartMicro is utilised to decode .asc file into a .csv file for further data analysis.

An additional algorithm was implemented by creating a pool of processes, also known as multiprocessing, that can (in principle) execute each sensor data acquisition process independently from each other in order to achieve simultaneous data acquisition. Pseudocode for simultaneously acquiring data is shown in Algorithm 2.

Algorithm 2 Summary of Simultaneous Sensor Data Acquisition

```

1: import ThreadPool from multiprocessing
2: initialize ThreadPool with 3 processes as pool
3: pool start camera data acquisition process
4: pool start lidar data acquisition process
5: pool start radar data acquisition process

```

A total of four trials were conducted to test for robustness of the proposed algorithms, in which the process was initiated and terminated via PyCharm IDE and the “taskkill” command in command line interface. Table I shows the duration of camera, lidar and radar data acquisition process including start time (first

data packet arrived) and end time (last data packet arrived) in the format of hh:mm:ss.ms. Despite the required synchronous start and end time, there is up to approximately seven seconds difference in the duration of data capture.

TABLE I. DURATION OF SENSORS DATA ACQUISITION PROCESS

		<i>Start Time</i>	<i>End Time</i>	<i>Duration</i>
<i>Trial 1</i>	Camera	16:34:36.476025	16:36:33.403071	00:01:56.927
	LiDAR	16:34:35.955142	16:36:36.544742	00:02:00.589
	Radar	16:34:35.568452	16:36:37.649943	00:02:02.081
<i>Trial 2</i>	Camera	11:20:44.773408	11:22:32.097489	00:01:47.324
	LiDAR	11:20:43.928155	11:22:38.512773	00:01:54.495
	LiDAR	11:20:43.528737	11:22:38.405229	00:01:54.876
<i>Trial 3</i>	Camera	13:03:43.214133	13:04:22.862680	00:00:39.649
	LiDAR	13:03:42.694605	13:04:22.924114	00:00:40.229
	Radar	13:03:42.286580	13:04:22.860784	00:00:40.574
<i>Trial 4</i>	Camera	13:37:18.807566	13:38:00.311562	00:00:41.504
	LiDAR	13:37:17.906356	13:38:00.930379	00:00:43.024
	Radar	13:37:17.500089	13:38:00.772766	00:00:43.273

The start time difference between radar, lidar and camera varies in each trial as shown in Table II. The average start time difference between camera, LiDAR and radar was calculated and the results show that LiDAR is, on average, 0.696 seconds ahead of the camera; the Radar is 1.096 second ahead of camera and 0.392 second ahead of the LiDAR. The results from the average start time difference demonstrated that the first frames or first data packet arriving at the endpoint is slowest for camera, and is fastest for radar sensor.

TABLE II. START TIME DIFFERENCE IN SECOND BETWEEN SENSORS

<i>Comparison</i>	<i>Which data arrive first at endpoint?</i>	<i>Trial 1</i>	<i>Trial 2</i>	<i>Trial 3</i>	<i>Trial 4</i>
Camera vs. LiDAR	LiDAR data	0.521s	0.845s	0.519s	0.901s
Camera vs. Radar	Radar data	0.908s	1.244s	0.927s	1.307s
LiDAR vs. Radar	Radar data	0.357s	0.399s	0.408s	0.406s

B. Discussion

The results shown in Table II indicate that the data acquired from each sensor were not synchronised with one another. Data from multiple sensors arriving with different and variable delays at the endpoint may cause inaccurate estimation of positions of the perceived objects, which is not acceptable especially in time and safety-critical tasks such as real-time collision avoidance. Further research is ongoing so as to minimise latencies that cannot be directly determined or repeatedly forecast. The current research programme is evaluating delays of each packet in the Medium Access Control (MAC) layer prior to the actual transmission or prior to the time spent in propagation of messages between network interfaces of sender and receiver [25].

Time Synchronisation is crucial in time critical multi-sensor fusion systems in order to accurately detect objects in real-time to prevent unnecessary collisions. Passive solutions like simplified drift-less version or drift model were proposed by Edwin Olson in 2010 [27]. Borgmann, B. et al., utilised an inertial navigation system (INS) to provide a central common clock for precise hardware synchronisation in their proposed multi-sensor vehicle [14]. This, and other techniques like synchronisation using an external UTC reference clock [8] are to be implemented and evaluated and will be reported in future papers.

V. CONCLUSION AND FUTURE WORK

In the present work, a sensor fusion system for collision avoidance in large, heavy off-road vehicles in an industrial environment was proposed. A Falcon-IQ IP industrial zoom camera, a Velodyne LiDAR Ultra Puck VLP-32C and a SmartMicro automotive radar UMRR-96 Type 153 sensor are utilized in our system. An algorithm to simultaneously collect data from camera, LiDAR and radar was evaluated and the results indicate that the start time for camera is the slowest and that the radar sensor is the fastest. The data collected from the three sensors were not synchronised with one another, and the challenges of time synchronisation in multi-sensor systems have been highlighted. Further review of the art and implementation of time synchronisation techniques described in previous works [25-27] will be conducted to minimise unknown latencies in multi-sensor fusion systems for real-time perception of the surroundings. This is essential to provide accurate estimation of the presence and position of objects in autonomous operation.

ACKNOWLEDGMENT

This paper and the research behind it would not have been possible without the support of the IMaR team in the Institute of Technology, Tralee. The authors are particularly grateful to Dr. Pat Doody for his introductions and ongoing support. Authors thank Denis O' Mahony and colleagues for their assistance in fabrication and machining; Anshul Awasthi and Krishna Panduru for their guidance on electrical wiring and electrical architecture, Torben Bruns of Smart Micro for his guidance on sourcing the suitable hardware interface for acquiring radar data and colleagues in the School of Science, Technology, Engineering and Maths in the Institute of Technology, Tralee for their support.

REFERENCES

- [1] U. P. Cambridge, 2020. *Cambridge Dictionary*. [Online]. Available at: <https://dictionary.cambridge.org/dictionary/english/autonomous>. [Accessed 12 March 2020]
- [2] Zoldy, M. & Zsombok, I., 2019. Influence of External Environmental Factors on Range Estimation of Autonomous Hybrid Vehicles. *CzOTO*, I(1), pp. 472-480.
- [3] Xue, et al., 2017. A vision-centered multi-sensor fusing approach to self-localization and obstacle perception for robotic cars. *Frontiers of Information Technology & Electronic Engineering*, XVIII(1), pp. 122-138.
- [4] Aeberhad, M. & Kaempchen, N., 2011. High-level sensor data fusion architecture for vehicle surround environment perception. *Proc. 8th Int. Workshop Intell. Transp.*
- [5] Tzafestas, S. G., 2014. *Introduction to Mobile Robot Control*. 1st ed. London: Elsevier.
- [6] Banerjee, K. et al., 2018. *Online Camera LiDAR Fusion and Object Detection on Hybrid Data for Autonomous Driving*. Suzhou, China, IEEE Intelligent Vehicles Symposium (IV), pp. 1632-1638.
- [7] Naidu, V., Gopalratnam, G., Raol, J. & Appavu, R. R., 2005. Data Association and Fusion Algorithms for Tracking in Precense of Measurement Loss. *IE (I) Journal-AS*, LXXXVI(1), pp. 564-575.
- [8] Westenberger, A. et al., 2011. *Temporal Synchronization in Multi-Sensor Fusion for Future Driver Assistance Systems*. Munich, IEEE International Symposium on Precision Clock Synchronization for Measurement, Control and Communication, pp. 93-98.
- [9] Huck, T. et al., 2011. *Precise Timestamping and Temporal Synchronization in Multi-Sensor Fusion*. Baden-Baden, 2011 IEEE Intelligent Vehicles Symposium (IV), pp. 242-247.
- [10] Kim, S., Kim, H., Yoo, W. & Huh, K., 2016. Sensor Fusion Algorithm Design in Detecting Vehicles Using Laser Scanner and Stereo Vision. *IEEE Transactions on Intelligent Transportation Systems*, XVII(4), pp. 1072-1084.
- [11] Mathas, C., 2013. *Sensor Data Fusion: More than Just Sensor Integration*. [Online]. Available at: <https://www.digikey.ie/en/articles/techzone/2013/apr/sensor-data-fusion-more-than-just-sensor-integration>. [Accessed 18 March 2020].
- [12] Jahromi, B. S., Tulabandhula, T. & Cetin, S., 2019. Real-Time Hybrid Multi-Sensor Fusion Framework for Perception in Autonomous Vehicles. *Sensors*, XIX(20), pp. 4357-4380.
- [13] CreativeLive, I., 2020. *How does a camera work?*. [Online]. Available at: <https://www.creativelive.com/photography-guides/how-does-a-camera-work>. [Accessed 19 March 2020].
- [14] Borgmann, B. et al., 2018. Data Processing and Recording Using a Versatile Multi-Sensor Vehicle. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, IV(1), pp. 21-28.
- [15] Outsight, 2019. *Fused Sensing vs. Sensor Fusion*. [Online]. Available at: <https://www.outsight.tech/technology/fused-sensing-vs.-sensor-fusion>. [Accessed 19 March 2020].
- [16] Clark, R., 2002. *Selecting a LIDAR System*. [Online]. Available at: <http://www.acuitylidar.com/PDFs/Selecting%20a%20LIDAR%20System.pdf>. [Accessed 19 March 2020].
- [17] Lu, J., Sibai, H., Fabry, E. & Forsyth, D., 2017, NO Need to Worry about Adversarial Examples in Object Detection in Autonomous Vehicles. *CoRR*, abs(1707.05301), pp. 1-9.
- [18] Kutila, M. et al., 2016. *Automotive LIDAR Sensor Development Scenarios for Harsh Weather Conditions*. Rio de Janeiro, IEEE.
- [19] Velodyne, n.d. *Velodyne VLP32C Data Sheet*. [Online] Available at: <https://velodynelidar.com/products/ultra-puck/>. [Accessed 20 March 2020].
- [20] Viatch, 2019. *Which Sensors are Best for Autonomous Vehicles? Cameras, Radar or Lidar?*. [Online]. Available at: <https://www.viatech.com/en/2019/09/which-sensors-are-best-for-autonomous-vehicles-cameras-radar-or-lidar>. [Accessed 20 March 2020].
- [21] Pulli, K., Baksheev, A., Korniyakov, K. & Eruhimov, V., 2012. Realtime Computer Vision with OpenCV. *Queue*, X(4), pp. 1-17.
- [22] Zelic, F. & Sable, A., 2020. *A comprehensive guide to OCR with Tesseract, OpenCV and Python*. [Online]. Available at: <https://nanonets.com/blog/ocr-with-tesseract>. [Accessed 21 March 2020].
- [23] Scapy, 2020. *Introduction: About Scapy*. [Online]. Available at: <https://scapy.readthedocs.io/en/latest/introduction.html>. [Accessed 21 March 2020].
- [24] Python-CAN, 2020. *Python-can*. [Online]. Available at: <https://python-can.readthedocs.io/en/master/>. [Accessed 21 March 2020].
- [25] Sivrikaya, F. & Yener, B., 2004. Time Synchronization in Sensor Networks: A Survey. *IEEE Network*, XVIII(4), pp.45-50.
- [26] nd.edu, 2020. *Chapter 9: Time Synchronisation*. [Online]. Available at: <https://www3.nd.edu/~cpoellab/teaching/cse40815/Chapter9.pdf>. [Accessed 22 March 2020].
- [27] Oslon, E., 2010. *A Passive Solution to the Sensor Synchronisation Problem*. Taipei, Taiwan, IEEE.
- [28] Kocić, J., Jovičić, N. & Drndarević, V., 2018. *Sensors and Sensor Fusion in Autonomous Vehicles*. Belgrade, IEEE.

- [29] Cattaneo, C., Mainetti, G. & Sala, R., 2015. The Importance of Camera Calibration and Distortion Correction to Obtain Measurements with Video Surveillance Systems. *Journal of Physics: Conference Series*, DCLVIII(1), pp. 1-7.
- [30] Joshi, P., 2014. *Understanding Camera Calibration*. [Online]. Available at: <https://prateekvjoshi.com/2014/05/31/understanding-camera-calibration/>. [Accessed 18 May 2020].